

Topics for Joint Project Incubation

— 4 $\frac{1}{2}$ Proposals —

Martin Fränzle (U Oldenburg, DE)

Mini-Symposium U Groningen, 14.9.2026

Common Motivation: Safety-Critical Human-Cyber-Physical Systems



Safety is a societal quest!

- Stringent guarantee of (esp.) safety properties required

Common Motivation: Safety-Critical Human-Cyber-Physical Systems



Safety is a societal quest!

- Stringent guarantee of (esp.) safety properties required,
- but any (static or dynamic) information inherently is inexact & partial!

Suggested Topics for Cooperation

Joint theme: Safe operation under imprecise observation of the situation.

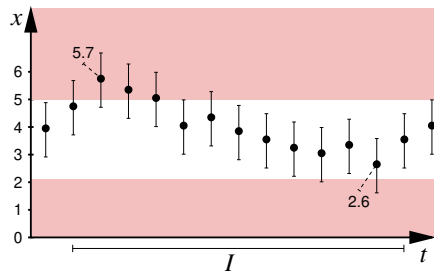
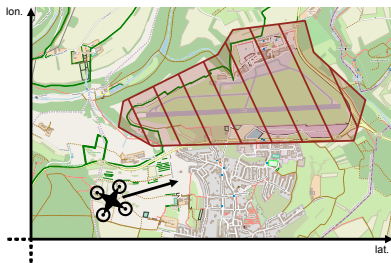
1. Game-theoretic synthesis of safe control under delayed state observation
— cf. previous meeting
2. Resilience of runtime verification / critical guard evaluation
 - 2.a Optimally precise monitoring verdicts when state or time observation is imprecise
 - 2.b Optimizing resilience of safety-critical guard evaluations against misperception
3. Contract-based design and analysis under uncertainty
4. A hyper-game semantics for SoS explanations and their generation

Suggested Topics for Cooperation

Joint theme: Safe operation under imprecise observation of the situation.

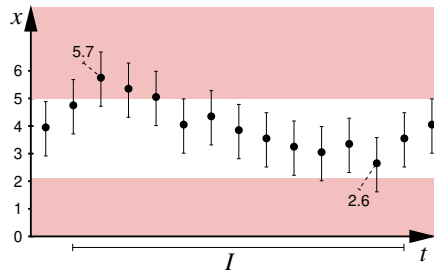
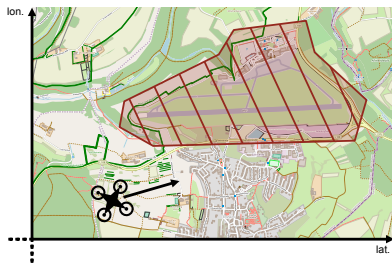
1. Game-theoretic synthesis of safe control under delayed state observation
— cf. previous meeting
2. Resilience of runtime verification / critical guard evaluation
 - 2.a Optimally precise monitoring verdicts when state or time observation is imprecise
 - 2.b Optimizing resilience of safety-critical guard evaluations against misperception
3. Contract-based design and analysis under uncertainty
4. A hyper-game semantics for SoS explanations and their generation

2.a Monitoring Under Uncertainty: The Problem



Safety condition: Avoid red area for next 12 time units: $G_{\leq 12}(x \geq 2 \wedge x \leq 5)$

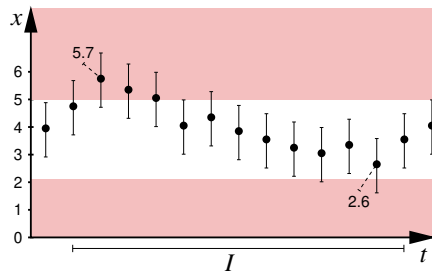
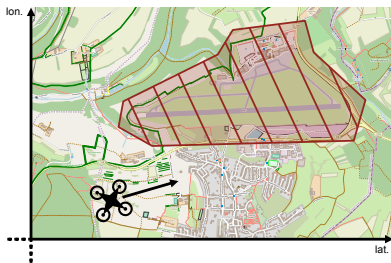
2.a Monitoring Under Uncertainty: The Problem



Safety condition: Avoid red area for next 12 time units: $G_{\leq 12}(x \geq 2 \wedge x \leq 5)$

- nominal measurements indicate violation,
- ground-truth intervals remain inconclusive.

2.a Monitoring Under Uncertainty: The Problem



Safety condition: Avoid red area for next 12 time units: $G_{\leq 12}(x \geq 2 \wedge x \leq 5)$

- nominal measurements indicate violation,
- ground-truth intervals remain inconclusive.

Ought we activate the safety mechanism?

Central Research Questions

- Given a durational safety property ϕ ,
- given a time series of measurements,
- given a mathematical characterization of measurement tolerances,

I. What does optimal monitoring, i.e. as exact as possible evaluation of ϕ over the time series mean?

Central Research Questions

- Given a durational safety property ϕ ,
 - given a time series of measurements,
 - given a mathematical characterization of measurement tolerances,
-
- I. What does optimal monitoring, i.e. as exact as possible evaluation of ϕ over the time series mean?
 - II. Can we algorithmically achieve such optimal monitoring?

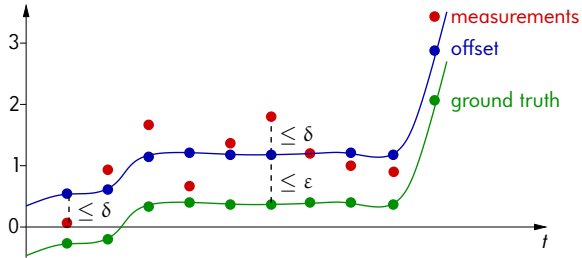
Central Research Questions

- Given a durational safety property ϕ ,
- given a time series of measurements,
- given a mathematical characterization of measurement tolerances,

- I. What does optimal monitoring, i.e. as exact as possible evaluation of ϕ over the time series mean?
- II. Can we algorithmically achieve such optimal monitoring?
- III. Is it inherently different from first pursuing optimal state estimation and then evaluating ϕ over the estimates?

The Model of Measurement Error: Trueness and Precision

[ISO 5725]



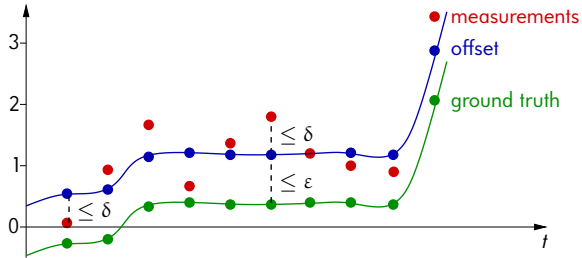
- Given some ground-truth trajectory τ , measuring time series m_s is possible iff

$$\exists o \in [-\varepsilon, \varepsilon] : \forall t \in T : \exists e \in [-\delta, \delta] : \tau(s)(t) + o + e = m_s(t), \quad (I)$$

where T is the set of time instants where measurements are taken.

The Model of Measurement Error: Trueness and Precision

[ISO 5725]



- Given some ground-truth trajectory τ , measuring time series m_s is possible iff

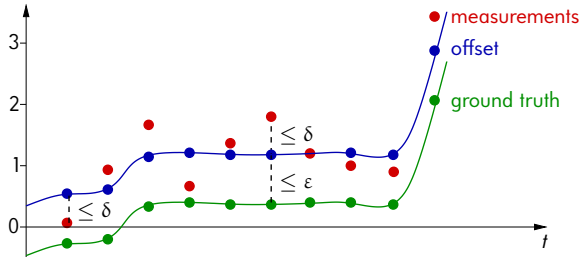
$$\exists o \in [-\varepsilon, \varepsilon] : \forall t \in T : \exists e \in [-\delta, \delta] : \tau(s)(t) + o + e = m_s(t), \quad (I)$$

where T is the set of time instants where measurements are taken.

- Vice versa, a ground truth τ is consistent with measurement series $m_{s_1}, \dots, m_{s_n}$, denoted by $m_{s_1}, \dots, m_{s_n} \vdash \tau$, iff all m_{s_i} are possible w.r.t. τ .

The Model of Measurement Error: Trueness and Precision

[ISO 5725]



- Given some ground-truth trajectory τ , measuring time series m_s is possible iff

$$\exists o \in [-\varepsilon, \varepsilon] : \forall t \in T : \exists e \in [-\delta, \delta] : \tau(s)(t) + o + e = m_s(t), \quad (I)$$

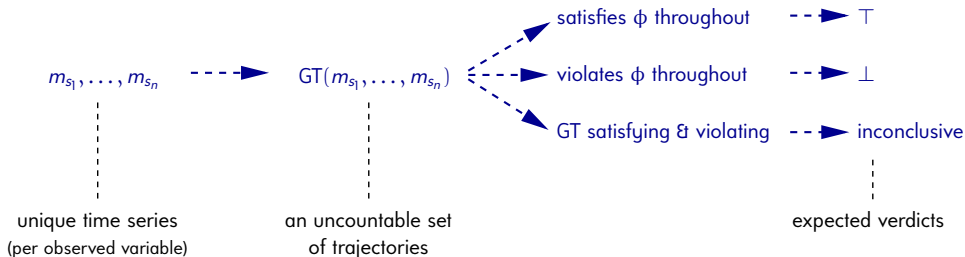
where T is the set of time instants where measurements are taken.

- Vice versa, a ground truth τ is consistent with measurement series $m_{s_1}, \dots, m_{s_n}$, denoted by $m_{s_1}, \dots, m_{s_n} \vdash \tau$, iff all m_{s_i} are possible w.r.t. τ .
- $GT(m_{s_1}, \dots, m_{s_n}) = \{\tau \mid m_{s_1}, \dots, m_{s_n} \vdash \tau\}$ is the possible ground truth given $m_{s_1}, \dots, m_{s_n}$.

The Monitoring Problem

is given some measurement time series

- to characterize the possible ground truth exactly and
- to determine truth values of the monitoring condition ϕ across the possible ground truth.



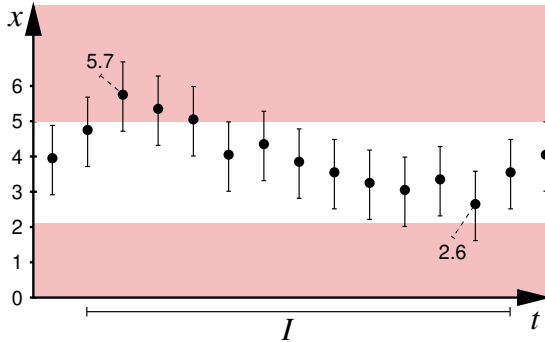
Properties of Monitoring Algorithms¹

- $\mathcal{M}$ is sound iff
 - (a) $\mathcal{M}(t, m_{S_1}, \dots, m_{S_n}) = \top$ implies that $\forall \tau \in \text{GT}(m_{S_1}, \dots, m_{S_n}) : \tau, t - \text{duration}(\phi) \models \phi$
and
 - (b) $\mathcal{M}(t, m_{S_1}, \dots, m_{S_n}) = \perp$ implies that $\forall \tau \in \text{GT}(m_{S_1}, \dots, m_{S_n}) : \tau, t - \text{duration}(\phi) \models \neg \phi$
- $\mathcal{M}$ is conclusive iff
 - (c) $\mathcal{M}(t, m_{S_1}, \dots, m_{S_n}) = \text{inconclusive}$ implies that

$$\exists \tau, \bar{\tau} \in \text{GT}(m_{S_1}, \dots, m_{S_n}) : \left(\begin{array}{l} \tau, t - \text{duration}(\phi) \models \phi \\ \bar{\tau}, t - \text{duration}(\phi) \models \neg \phi \end{array} \wedge \right)$$
- $\mathcal{M}$ is complete iff $\mathcal{M}$ terminates on all t and m_{S_i} .
- $\mathcal{M}$ is exact iff $\mathcal{M}$ is sound, conclusive, and complete.

¹B Finkbeiner, MF, F Kohn, P Kröger: *A truly robust signal temporal logic: Monitoring safety properties of interacting cyber-physical systems under uncertain observation*, Algorithms 15(4)

Exact Monitoring in the Model-Free Case

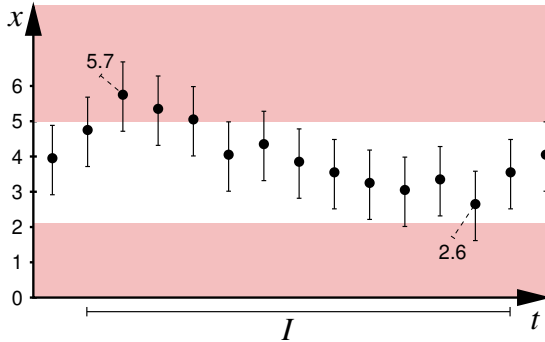


- Offset error bound $\varepsilon = 0.5$
- Per-sample error bound $\delta = 0.5$
- **Task:** decide at time $t = 13$ whether

$$G_{\leq 12}(x \geq 2 \wedge x \leq 5)$$

held recently.

Exact Monitoring in the Model-Free Case



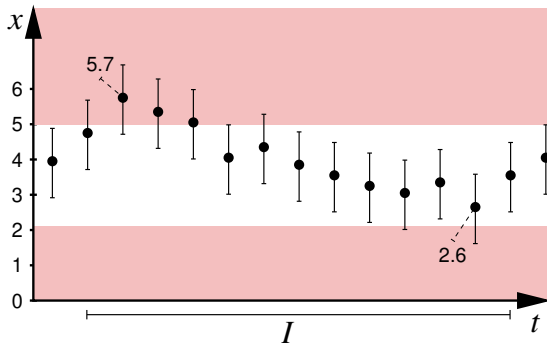
- Offset error bound $\varepsilon = 0.5$
- Per-sample error bound $\delta = 0.5$
- **Task:** decide at time $t = 13$ whether

$$G_{\leq 12}(x \geq 2 \wedge x \leq 5)$$

held recently.

- Remains inconclusive under (best possible) state estimation;
- In fact definitely is $\perp$.

Exact Monitoring in the Model-Free Case

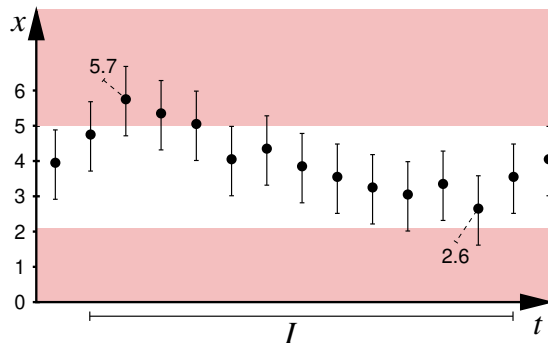


- Offset error bound $\varepsilon = 0.5$
- Per-sample error bound $\delta = 0.5$
- **Task:** decide at time $t = 13$ whether

$$G_{\leq 12}(x \geq 2 \wedge x \leq 5)$$

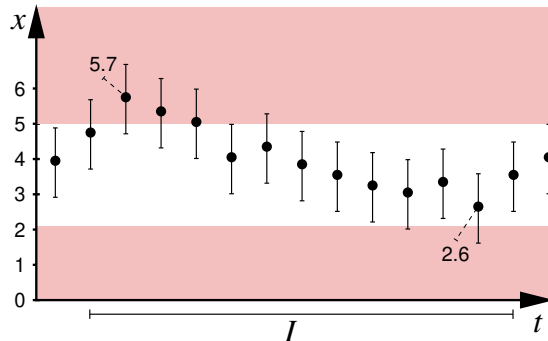
held recently.

- Remains inconclusive under (best possible) state estimation;
- In fact definitely is $\perp$.
- Encoding as Boolean constraint system over affine forms possible; SMT(LA) can decide the problem.



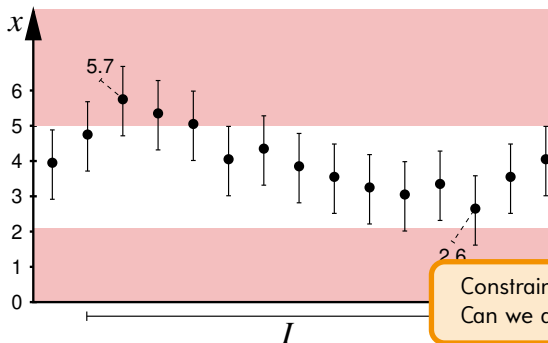
- Condition: $\psi = \dots \wedge x_3 + o + e_3 = m_3 \wedge \dots \wedge x_{12} + o + e_{12} = m_{12} \wedge \dots$
 $\wedge o, \dots, e_3, \dots, e_{12}, \dots \in [-0.5, 0.5]$
 $\wedge \dots \wedge m_3 = 5.7 \wedge \dots \wedge m_{12} = 2.6 \wedge \dots \wedge$
 $\wedge \dots, x_3, \dots, x_{12}, \dots \in [2, 5]$
- Unsatisfiable, decided by SMT(LA)

E.g., Condition for Possibility of Property Violation [Finkbeiner, MF, Kohn, Kröger '22]



- Condition: $\psi = \dots \wedge x_3 + o + e_3 = m_3 \wedge \dots \wedge x_{12} + o + e_{12} = m_{12} \wedge \dots$
 $\wedge o, \dots, e_3, \dots, e_{12}, \dots \in [-0.5, 0.5]$
 $\wedge \dots \wedge m_3 = 5.7 \wedge \dots \wedge m_{12} = 2.6 \wedge \dots \wedge$
 $\wedge \dots, x_3, \dots, x_{12}, \dots \in [2, 5]$
- Unsatisfiable, decided by SMT(LA) $\Rightarrow$ Monitoring verdict is a definite $\perp$.

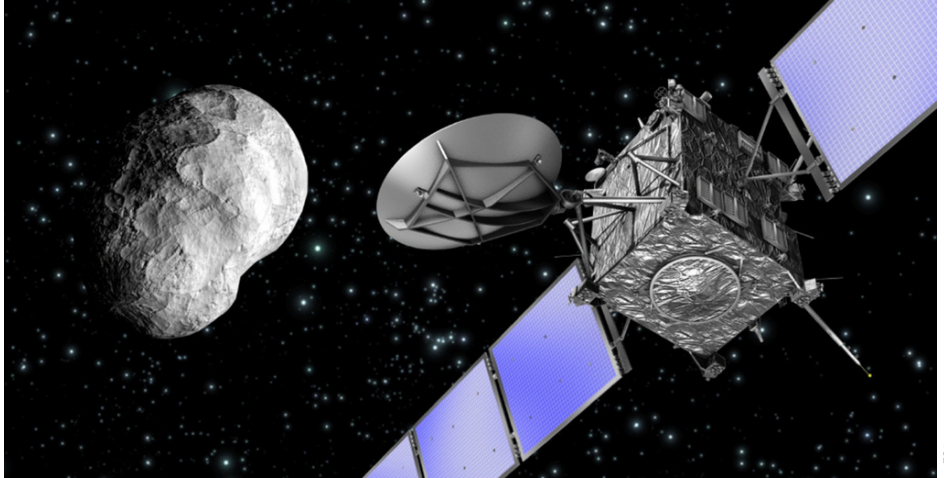
E.g., Condition for Possibility of Property Violation [Finkbeiner, MF, Kohn, Kröger '22]



Constraint systems grows linearly over time!
Can we do with bounded size?

- Condition: $\psi = \dots \wedge x_3 + o + e_3 = m_3 \wedge \dots \wedge x_{12} + o + e_{12} = m_{12} \wedge \dots$
 $\wedge o, \dots, e_3, \dots, e_{12}, \dots \in [-0.5, 0.5]$
 $\wedge \dots \wedge m_3 = 5.7 \wedge \dots \wedge m_{12} = 2.6 \wedge \dots \wedge$
 $\wedge \dots, x_3, \dots, x_{12}, \dots \in [2, 5]$
- Unsatisfiable, decided by SMT(LA) $\Rightarrow$ Monitoring verdict is a definite $\perp$.

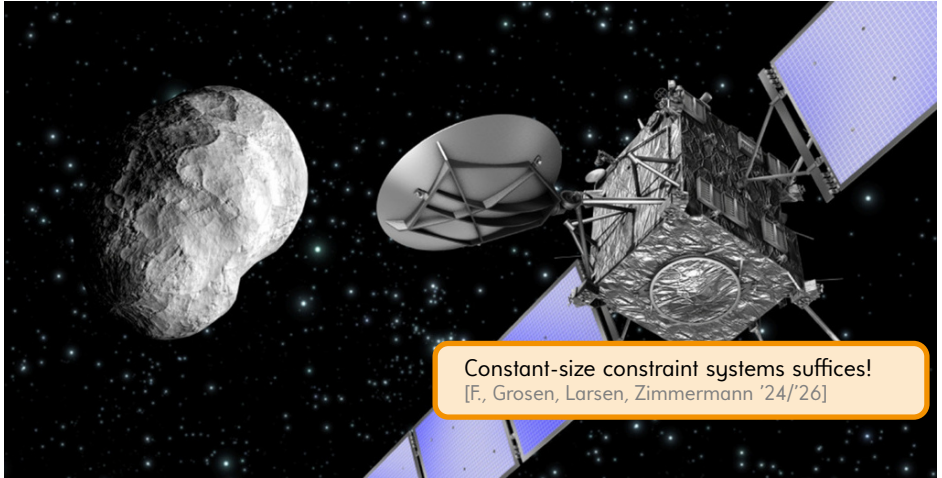
Problem Variant: Dense Time & Delayed Observation



© ESA

Task: Monitor a dense-time temporal property of a process of interest via a delaying channel.

Problem Variant: Dense Time & Delayed Observation



© ESA

Task: Monitor a dense-time temporal property of a process of interest via a delaying channel.

Discussion & Future Work

Result: Optimal (“exact”) monitoring procedures for temporal properties under uncertain observation

- sound, never give a wrong verdict despite uncertainty,
- as conclusive as possible given the uncertainty,
- cover inexact and/or intermittent state/time observation, where “inexact” refers to bounded offset & per-sample errors.

Key insight: Optimal monitoring $\neq$ Optimal state estimation + Property evaluation

Discussion & Future Work

Result: Optimal (“exact”) monitoring procedures for temporal properties under uncertain observation

- sound, never give a wrong verdict despite uncertainty,
- as conclusive as possible given the uncertainty,
- cover inexact and/or intermittent state/time observation, where “inexact” refers to bounded offset & per-sample errors.

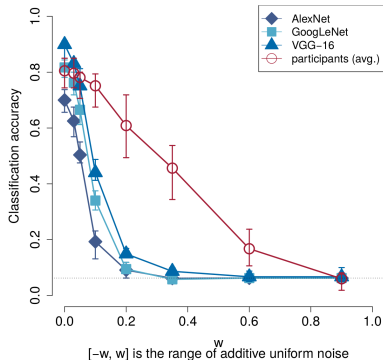
Key insight: Optimal monitoring $\neq$ Optimal state estimation + Property evaluation

Future work / Possible joint projects:

- Combine state and timing uncertainties within a single model & integrated algorithm.
- Extension to stochastic model of per-sample measurement error & offset.
- Characterize property classes for which exactness can/cannot be achieved by bounded-size (and thus bounded-dimensional) CSPs.

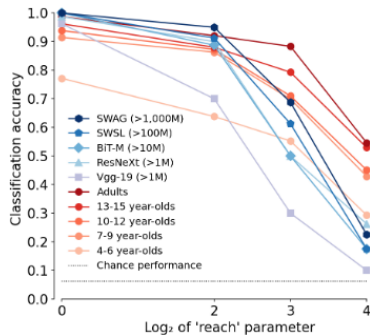
2.b Optimizing Resilience of Critical Guard Evaluation: The Problem

— Environmental perception is orders of magnitude less reliable than our safety targets!



Classification accuracy under unif. additive noise

[Geirhos, Janssen, Schütt et al. '17]²



Classification accuracy under "eidolon" distortion

[Huber, Geirhos, Wichmann '21]³

²R. Geirhos, D.H.J. Janssen, H.H. Schütt, J. Rauber, M. Bethge, F.A. Wichmann: *Comparing deep neural networks against humans: object recognition when the signal gets weaker*. <https://arxiv.org/pdf/1706.06969.pdf>, 2017

³L.S. Huber, R. Geirhos, F.A. Wichmann: *The developmental trajectory of object recognition robustness: children are like small adults but unlike big deep neural networks*. <https://arxiv.org/pdf/2205.10144.pdf>, 2021]

Architecture-Ctrl.ed Safety Impact: Functional (Ir-)Relevance of Percepts



Architecture-Ctrl.ed Safety Impact: Functional (Ir-)Relevance of Percepts



Architecture-Ctrl.ed Safety Impact: Functional (Ir-)Relevance of Percepts



Architecture-Ctrl.ed Safety Impact: Functional (Ir-)Relevance of Percepts



Architecture-Ctrl.ed Safety Impact: Functional (Ir-)Relevance of Percepts



Architecture-Ctrl.ed Safety Impact: Functional (Ir-)Relevance of Percepts



Functional impact, and thus impact on safety, of (mis-)percepts varies drastically!

Start Point: Guard Conditions for Critical Maneuvers

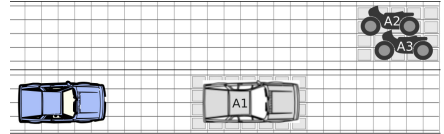
- due to supervisory control, shielding, Simplex architecture, ...

(Simplified) guard condition for overtaking:

$$guard \equiv necessary \wedge safe$$

$$necessary \equiv \bigvee_{x=6}^8 \bigvee_{y=1}^5 obstacle@(x_{ego} + x, y)$$

$$safe \equiv \neg \bigvee_{y=6}^{10} \left(\begin{array}{l} \bigvee_{x=1}^{20} pedestrian@(x_{ego} + x, y) \\ \bigvee_{x=-4}^{40} car@(x_{ego} + x, y) \\ \bigvee_{x=-1}^{60} motorcycle@(x_{ego} + x, y) \end{array} \quad \bigvee \right)$$



Start Point: Guard Conditions for Critical Maneuvers

– due to supervisory control, shielding, Simplex architecture, ...

(Simplified) guard condition for overtaking:

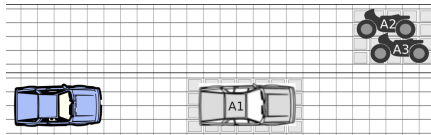
$$guard \equiv necessary \wedge safe$$

$$necessary \equiv \bigvee_{x=6}^8 \bigvee_{y=1}^5 obstacle@(x_{ego} + x, y)$$

massive disjunction of classifier outputs

$$safe \equiv \neg \bigvee_{y=6}^{10} \left(\begin{array}{l} \bigvee_{x=1}^{20} pedestrian@(x_{ego} + x, y) \\ \bigvee_{x=-4}^{40} car@(x_{ego} + x, y) \\ \bigvee_{x=-1}^{60} motorcycle@(x_{ego} + x, y) \end{array} \bigvee \right)$$

negation thereof



The Goal

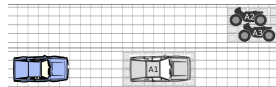
Given

- Boolean or three-valued output $\{t, \perp, f\}$ of classifiers for each grid element (x, y) and each object class C ,
- true and false positive rates $tp_{(x,y)}^C$ and $fp_{(x,y)}^C$ for each such atomic sensor,
- true and false negative rates $tn_{(x,y)}^C$ and $fn_{(x,y)}^C$ for each such atomic sensor,

enforce

Safety: the false positive rate $fp(guard)$ remains below a societally accepted risk threshold θ ,

Performance: the true positive rate $tp(guard)$ be maximal.



Observation: *There's Liberty in Phrasing *safe* and *necessary**

- The real world provides loads of invariants.
- E.g., any car occupies 6 grid elements at least.
- Under this invariant

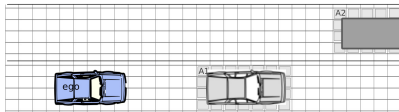
$$\phi \equiv \bigvee_{x=6}^8 \bigvee_{y=1}^5 \text{car}@(\mathbf{x}_{\text{ego}} + \mathbf{x}, \mathbf{y})$$

and

$$\phi_k \equiv \sum_{x=6}^8 \sum_{y=1}^5 \chi(\text{car}@(\mathbf{x}_{\text{ego}} + \mathbf{x}, \mathbf{y})) \geq k$$

are **logically equivalent** for any $k \in \{1, \dots, 6\}$
over ground truth.

- But their (true & false) **positive rates vary**
when interpreting noisy percepts.



Observation: *There's Liberty in Phrasing $safe$ and $necessary$*

- The real world provides loads of invariants.
- E.g., any car occupies 6 grid elements at least.
- Under this invariant

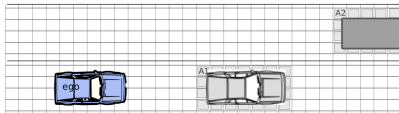
$$\phi \equiv \bigvee_{x=6}^8 \bigvee_{y=1}^5 car@(x_{ego} + x, y)$$

and

$$\phi_k \equiv \sum_{x=6}^8 \sum_{y=1}^5 \chi(car@(x_{ego} + x, y)) \geq k$$

are **logically equivalent** for any $k \in \{1, \dots, 6\}$
over ground truth.

- But their (true & false) **positive rates vary**
when interpreting noisy percepts.



We call such formulae W-equivalent.

W-Equivalent Formula Rewriting:

[MF, Hein '24][MF, Kröger, Nienaber '26]

— Results for $tp = 0.85$, $fp = 0.2$, $tn = 0.8$, $fn = 0.15$

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	
No c.	0.965	0.833	0.602	0.352	0.164	0.061	$fp(\Phi_k)$
1 c.	$1 - 2 \cdot 10^{-6}$	$1 - 6 \cdot 10^{-4}$	0.999	0.992	0.954	0.834	$tp(\Phi_k)$
2 c.	$1 - 6 \cdot 10^{-11}$	$1 - 5 \cdot 10^{-9}$	$1 - 1 \cdot 10^{-7}$	$1 - 3 \cdot 10^{-6}$	$1 - 4 \cdot 10^{-5}$	$1 - 4 \cdot 10^{-4}$	$tp(\Phi_k)$

W-Equivalent Formula Rewriting:

[MF, Hein '24][MF, Kröger, Nienaber '26]

— Results for $tp = 0.85$, $fp = 0.2$, $tn = 0.8$, $fn = 0.15$

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	
No c.	0.965	0.833	0.602	0.352	0.164	0.061	$fp(\phi_k)$
1 c.	$1 - 2 \cdot 10^{-6}$	$1 - 6 \cdot 10^{-4}$	0.999	0.992	0.954	0.834	$tp(\phi_k)$
2 c.	$1 - 6 \cdot 10^{-11}$	$1 - 5 \cdot 10^{-9}$	$1 - 1 \cdot 10^{-7}$	$1 - 3 \cdot 10^{-6}$	$1 - 4 \cdot 10^{-5}$	$1 - 4 \cdot 10^{-4}$	$tp(\phi_k)$

- W-equivalent formula ϕ_6 **reduces the safety-critical false positive rate ≈ 16 -fold**,
 - $\phi = \phi_1$ features ≈ 5 -fold fp of the atomic sensor,
 - ϕ_6 features just $\approx \frac{3}{10} fp$ of the atomic sensor,
- while maintaining true positive rate (i.e., performance) at the sensor performance level.

W-Equivalent Formula Rewriting:

[MF, Hein '24][MF, Kröger, Nienaber '26]

— Results for $tp = 0.85$, $fp = 0.2$, $tn = 0.8$, $fn = 0.15$

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	
No c.	0.965	0.833	0.602	0.352	0.164	0.061	$fp(\phi_k)$
1 c.	$1 - 2 \cdot 10^{-6}$	$1 - 6 \cdot 10^{-4}$	0.999	0.992	0.954	0.834	$tp(\phi_k)$
2 c.	$1 - 6 \cdot 10^{-11}$	$1 - 5 \cdot 10^{-9}$	$1 - 1 \cdot 10^{-7}$	$1 - 3 \cdot 10^{-6}$	$1 - 4 \cdot 10^{-5}$	$1 - 4 \cdot 10^{-4}$	$tp(\phi_k)$

- W-equivalent formula ϕ_6 **reduces the safety-critical false positive rate ≈ 16 -fold**,
 - $\phi = \phi_1$ features ≈ 5 -fold fp of the atomic sensor,
 - ϕ_6 features just $\approx \frac{3}{10} fp$ of the atomic sensor,
- while maintaining true positive rate (i.e., performance) at the sensor performance level.
- Figures even get **exponentially better over a (realistic) finer grid!**

Discussion

Approach: Don't-care-equivalent rewriting optimizes safety and performance of safety-critical autonomous systems

- by manipulating guard conditions (safeguards, guardrails, shields, ...) of critical maneuvers,
- preserving their logical content,
- yet optimizing resilience against misperception.

Discussion

Approach: Don't-care-equivalent rewriting optimizes safety and performance of safety-critical autonomous systems

- by manipulating guard conditions (safeguards, guardrails, shields, ...) of critical maneuvers,
- preserving their logical content,
- yet optimizing resilience against misperception.

Result: Safety gains by orders of magnitude have been observed

- on artificially small grid worlds ("toy examples"),
- but large (i.e., realistic) grids yield exponentially larger gains even.

Discussion

Approach: Don't-care-equivalent rewriting optimizes safety and performance of safety-critical autonomous systems

- by manipulating guard conditions (safeguards, guardrails, shields, ...) of critical maneuvers,
- preserving their logical content,
- yet optimizing resilience against misperception.

Result: Safety gains by orders of magnitude have been observed

- on artificially small grid worlds ("toy examples"),
- but large (i.e., realistic) grids yield exponentially larger gains even.

Future work / Possible joint projects: Optimal property rewriting is algorithmically extremely hard;

- develop approximate algorithms (with guaranties!) and appropriate data structures,
- identify formula structures and structural parameters characterizing tractable subproblems.